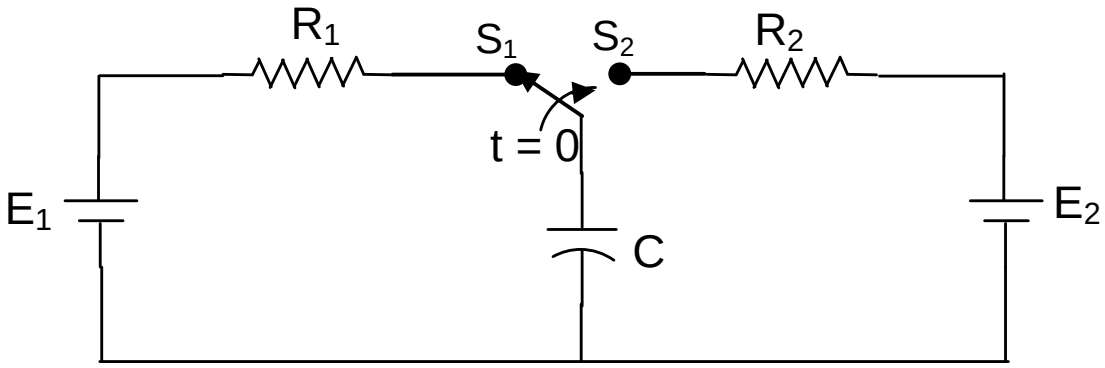
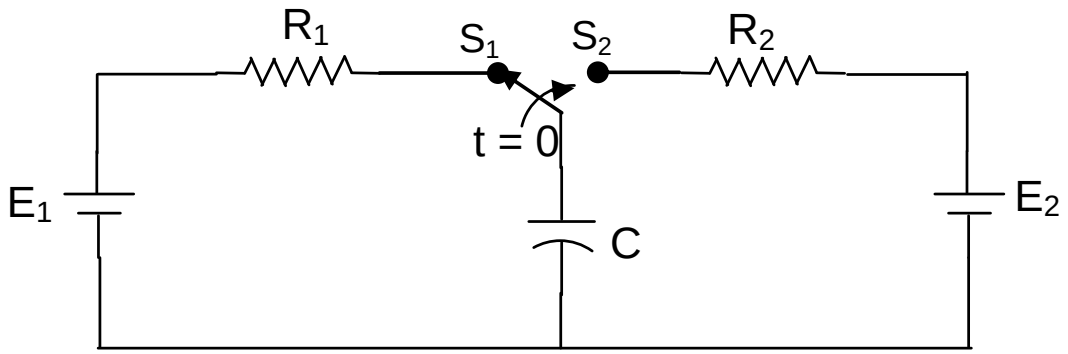
More general case of finding the capacitor voltage

In the previous discussion, it was assumed that the initial capacitor voltage $v_C(0) = 0$. There may be very many situations wherein initial capacitor voltage is not zero. There may be initial charge in the capacitor resulting non-zero initial capacitor voltage (Example 7.8). Further, the circuit arrangements can also cause non-zero initial capacitor voltage. For this purpose consider the circuit shown below. The switch was in position S_1 for a long time. It is moved from position S_1 to S_2 at time t = 0.





We shall assume the following:

1. At time $t = 0^-$ the circuit was at steady state condition with the switch in position S_1

2. After switching to position S_2, the circuit is allowed to reach the steady state condition

Thus, we are interested about the transient analysis for one switching period only.

Initial capacitor voltage $v_C(0)$ is E_1 and the final capacitor voltage $v_C(\infty)$,will be E_2.

The more general expression for the capacitor voltage can be obtained as

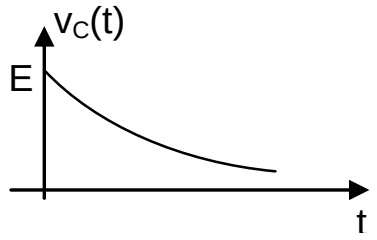
$$v_C(t) = v_C(\infty) + [v_C(0) - v_C(\infty)]\ e^{-\frac{1}{R_2 C} t} \qquad (7.47)$$

Summary of formulae useful for transient analysis on RC circuits

1. Time constant $\tau = RC$ $\quad \alpha = 1 / RC$

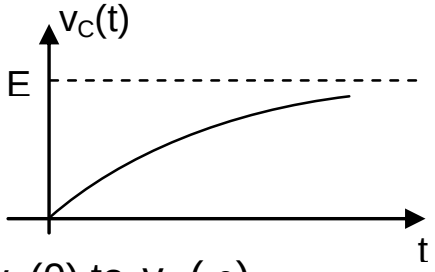
2. When the capacitor is discharging from the initial voltage of E

$$v_C(t) = E\, e^{-\frac{1}{RC}t}$$



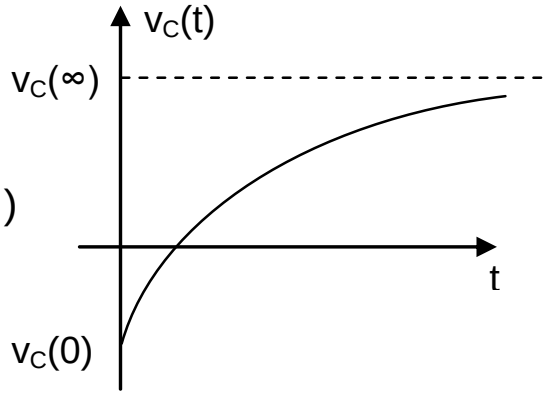
3. When the capacitor is charged from zero initial voltage to final voltage of E

$$v_C(t) = E\,(1 - e^{-\frac{1}{RC}t})$$



4. When the capacitor voltage changes from $v_C(0)$ to $v_C(\infty)$

$$v_C(t) = v_C(\infty) + [v_C(0) - v_C(\infty)]\, e^{-\frac{1}{RC}t}$$



Plot of $v_C(t)$ depends on values of $v_C(0)$ and $v_C(\infty)$

5. Capacitor current $i_C(t) = C\dfrac{dv_C(t)}{dt}$

Example 7.1 An RC circuit has R = 20 Ω and C = 400 μF. What is its time constant?

Solution For RC circuit, time constant τ = RC.

Therefore, $\tau = 20 \times 400 \times 10^{-6}$ s = 8 ms

Example 7.2 A capacitor in an RC circuit with R = 25 Ω and C = 50 μF is being charged with initial zero voltage. What is the time taken for the capacitor voltage to reach 40 % of its steady state value?

Solution With R = 25 Ω and C = 50 μF, τ = RC = 1.25×10^{-3} s; hence 1/RC = 800 s^{-1}.

Taking the capacitor steady state voltage as E, $v_C(t) = E\,(1 - e^{-\frac{1}{RC}t})$

Let t_1 be the time at which the capacitor voltage becomes 0.4 E. Then

$0.4\,E = E\,(1 - e^{-800\,t_1})$ i.e. $0.4 = 1 - e^{-800\,t_1}$

$e^{-800\,t_1} = 0.6$ i.e. $-800\,t_1 = \ln 0.6 = -0.5108$

Therefore, $t_1 = \frac{0.5108}{800}\text{s} = 0.6385 \times 10^{-3}$ s $= 0.6385$ ms

Example 7.3 In an RC circuit, having a time constant of 2.5 ms, the capacitor discharges with initial voltage of 80 V. (a) Find the time at which the capacitor voltage reaches 55 V, 30 V and 10 V (b) Calculate the capacitor voltage at time 1.2 ms, 3 ms and 8 ms.

Solution (a) Time constant RC = 2.5 ms; Thus $\frac{1}{RC} = \frac{1000}{2.5} = 400\ s^{-1}$

During discharge, capacitor voltage is given by $v_C(t) = 80\ e^{-400\,t}\ V$

Let t_1, t_2 and t_3 be the time at which capacitor voltage becomes 55 V, 30 V and 10 V.

$$55 = 80\,e^{-400 t_1};\ -400\ t_1 = \ln\frac{55}{80} = -0.3747;\ \text{Thus } t_1 = 0.93765\ \text{ms}$$

$$30 = 80\ e^{-400 t_2};\ -400\ t_2 = \ln\frac{30}{80} = -0.9808;\ \text{Thus } t_2 = 2.452\ \text{ms}$$

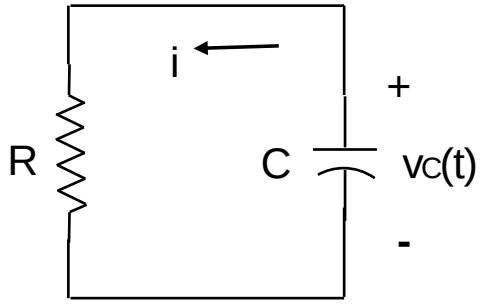
$$10 = 80\ e^{-400 t_3};\ -400\ t_3 = \ln\frac{10}{80} = -2.0794;\ \text{Thus } t_3 = 5.1985\ \text{ms}$$

(b) $v_C(1.2\text{x } 10^{-3}) = 80\ e^{-400\,t} = 80\ e^{-0.48} = 49.5027\ V$

$v_C(3\text{x } 10^{-3}) = 80\ e^{-400\,t} = 80\ e^{-1.2} = 24.0955\ V$

$v_C(8\text{x } 10^{-3}) = 80\ e^{-400\,t} = 80\ e^{-3.2} = 3.261\ V$

Example 7.4 Consider the circuit shown below.



Given

$v_C(t) = 56\, e^{-250\,t}$ V for $t > 0$

$i(t) = 7\, e^{-250\,t}$ mA for $t > 0$

(a) Find the values of R and C. (b) Determine the time constant.

(c) At what time the voltage $v_C(t)$ will reach half of its initial value?

Solution (a) Given that $v_C(t) = 56\, e^{-250\,t}$ V. Therefore $\tau = RC = \frac{1}{250}$ s

Resistance $R = \frac{v_C(t)}{i(t)} = 8000\,\Omega$; Thus capacitance $C = \frac{1}{250 \text{ X } 8000}$ F $= 0.5\,\mu$F

(b) Time constant = RC = 4 x 10^{-3} s = 4 ms

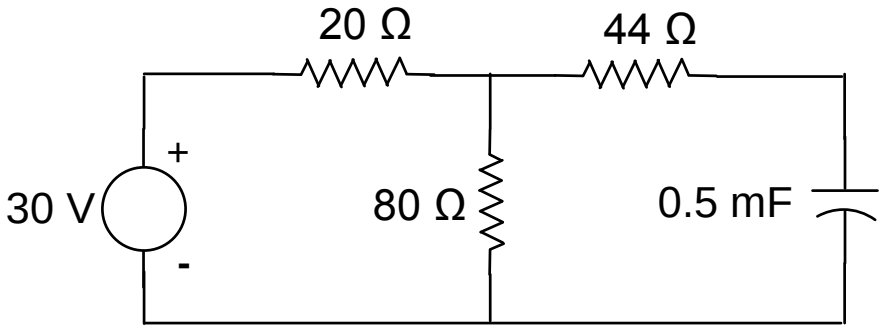
(c) Let t_1 be the time taken for the voltage to reach half of its initial value of 56 V.

Then, $56\, e^{-250 t_1} = 28$; i.e. $e^{-250 t_1} = 0.5$ i.e. - 250 t_1 = ln 0.5 = - 0.6931;

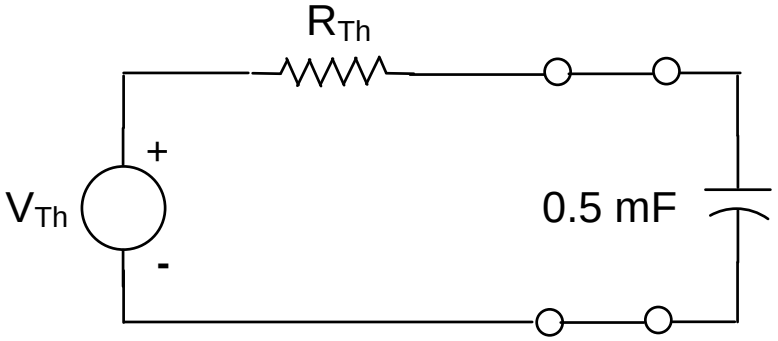
$$\text{Time } t_1 = \frac{0.6931}{250} \text{s} = 2.7724 \text{ x } 10^{-3} \text{ s} = 2.7724 \text{ ms}$$

Example 7.5

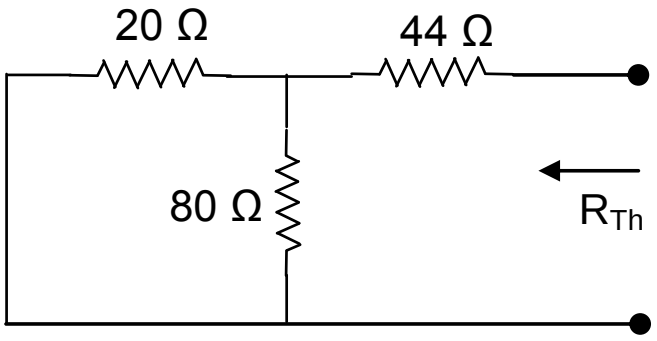
Find the time constant of the RC circuit shown in below.



Solution Thevenin's equivalent across the capacitor, is shown below.



(a)

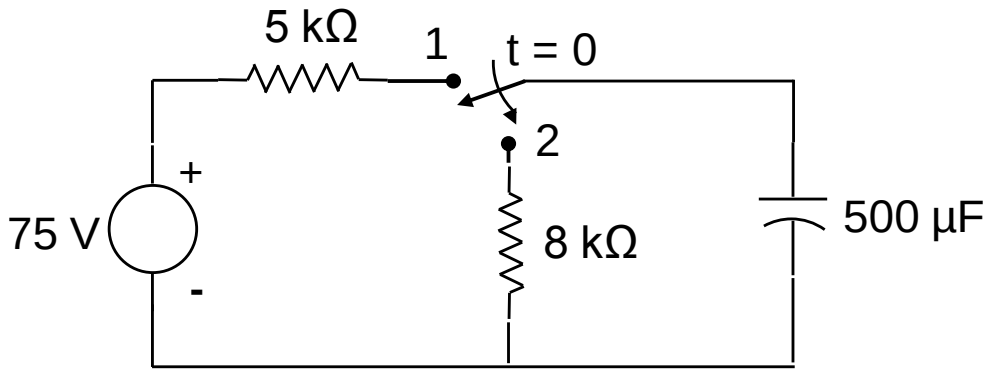


(b)

Referring to Fig. (b) above, $R_{Th} = 44 + (20 \,||\, 80) = 60\ \Omega$

Time constant $\tau = RC = 60 \times 0.5 \times 10^{-3}$ s = 30 ms

Example 7.6 The switch in circuit shown was in position1 for a long time. It is moved from position 1 to position 2 at time t = 0. Sketch the wave form of $v_C(t)$ for t > 0.



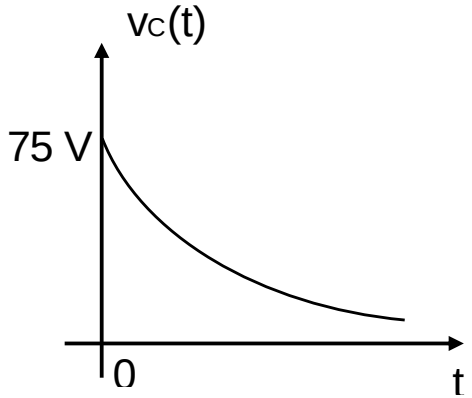
Solution With switch is in position 1, capacitor gets charged to a voltage of 75 V. i.e. $v_C(0^+) = 75$ V. The switch is moved to position 2 at time t = 0.

Time constant $RC = 8 \times 10^3 \times 500 \times 10^{-6} = 4$ s

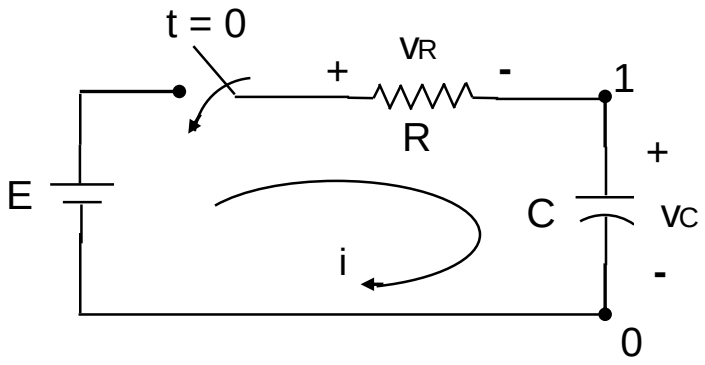
Finally the capacitor voltage decays to zero. Thus,

$$v_C(t) = 75\, e^{-0.25\, t}$$

Wave form of the capacitor voltage is shown.



Example 7.7 A series RC circuit has a constant voltage of E, applied at time t = 0 as shown in Fig. below. The capacitor has no initial charge. Find the equations for i, v_R and v_C. Sketch the wave shapes.



E = 100 V

R = 5000 Ω

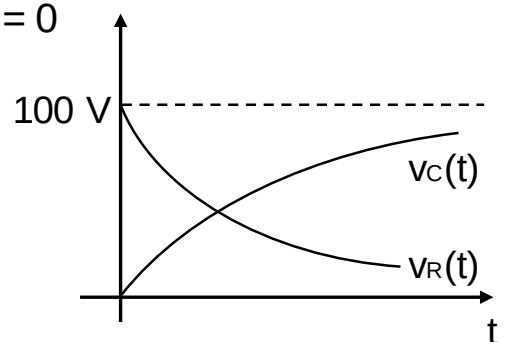
C = 20 μF

Solution Since there is no initial charge, $v_C(0^+) = v_C(0^-) = 0$

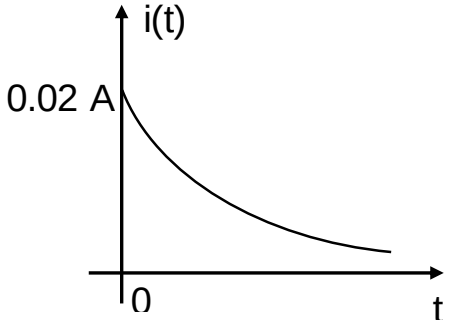
For t > 0, capacitor is charged to final voltage of 100 V.

Time constant RC = 5000 x 20 x 10^{-6} = 0.1 sec.

$v_C(t) = E\left(1 - e^{-\frac{1}{RC}t}\right)$. Thus, $v_C(t) = 100\left(1 - e^{-10\,t}\right)$ V



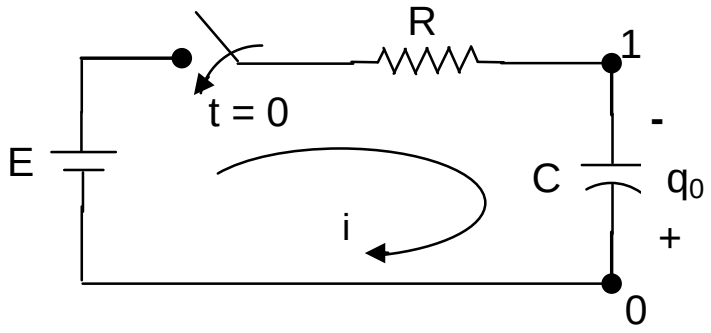
$$i(t) = C\frac{dv_C}{dt} = 20 \times 10^{-6} \times 100 \times 10\, e^{-10\,t} = 0.02\, e^{-10\,t}\ A$$



Voltage across the resistor is $v_R(t) = R\,i(t) = 100\, e^{-10\,t}$ V

Wave shapes of i, $v_R(t)$ and $v_C(t)$ are shown.

Example 7.8 A 20 μF capacitor in the RC circuit shown has an initial charge of q_0 = 500 μC with the polarity as shown. The switch is closed at time t = 0. Find the current transient and the voltage across the capacitor. Find the time at which the capacitor voltage is zero. Also sketch their wave shape.



E = 50 V

R = 1000 Ω

C = 20 μF

Solution Initial charge of q_0 in the capacitor is equivalent to initial voltage of

$$v_C(0) = -\frac{q_0}{C} = -\frac{500 \text{ X } 10^{-6}}{20 \text{ X } 10^{-6}} = -25 \text{ V};$$ Further, $v_C(\infty) = E = 50$ V

Time constant RC = 1000 X 20 X 10^{-6} = 20 X 10^{-3} s. Thus 1/RC = 50 s^{-1}

$$v_C(t) = v_C(\infty) + [v_C(0) - v_C(\infty)]\ e^{-\frac{1}{RC}t}$$

$$v_C(t) = 50 + [\,-25 - 50\,]\ e^{-50\,t} = 50 - 75\ e^{-50\,t}$$

$$\text{Current } i(t) = C\frac{dv_C}{dt} = 20 \text{ X } 10^{-6} \text{ X } 75 \text{ X } 50\ e^{-50\,t} \text{ A} = 0.075\ e^{-50\,t} \text{ A}$$

Let t_1 be the time at which the capacitor voltage becomes zero. Then

$50 - 75\ e^{-50\ t_1} = 0$ i.e. $e^{-50\ t_1} = 0.6667$

$-50\ t_1 = -0.4054$ i.e. $t_1 = 8.108 \times 10^{-3}$ s

The capacitor voltage becomes zero at time $t_1 = 8.108$ ms

Wave forms are shown in Fig. 7.24

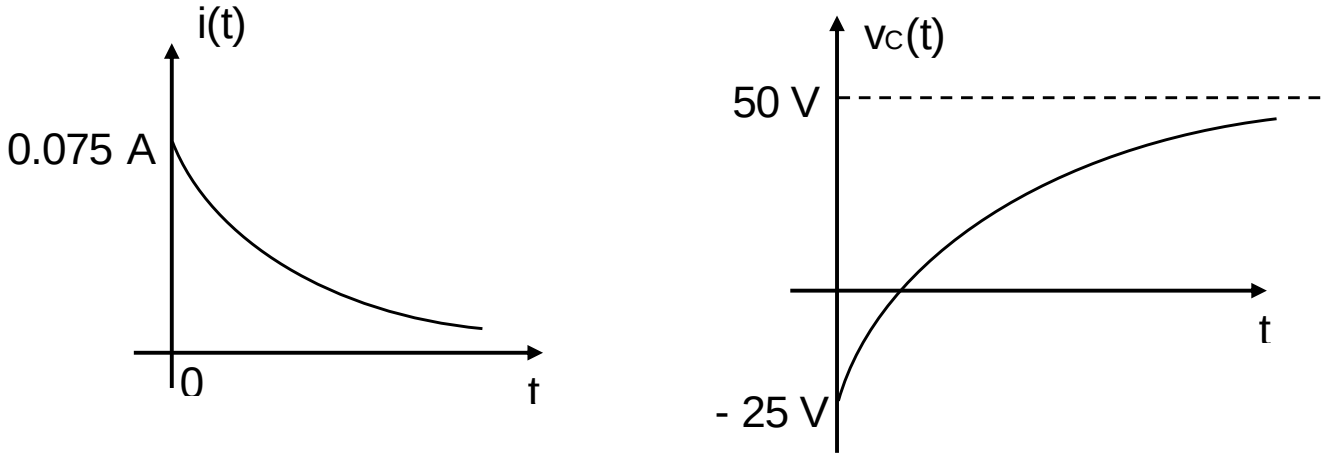
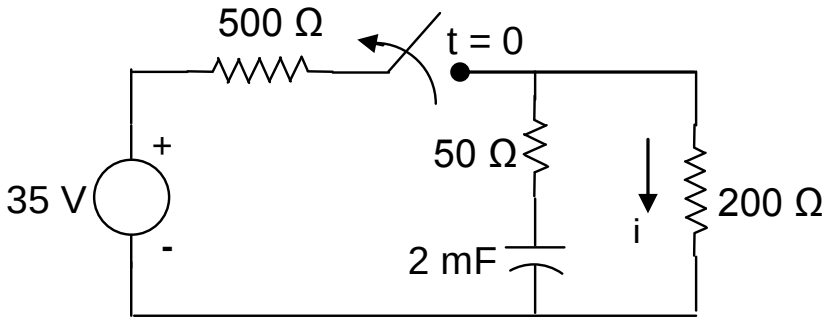


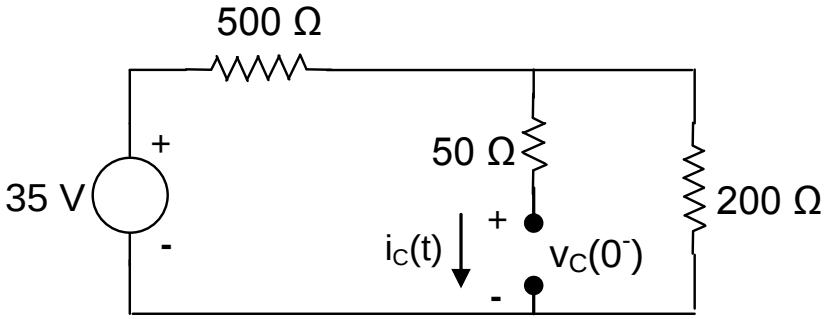
Fig. 7.24 Wave forms - Example 7.8.

Example 7.9

Consider the circuit shown below. The switch was in closed position for a long time. It is opened at time t = 0. Find the current i(t) for t > 0.



Solution Circuit at time t = 0 $^-$ is shown.



$$v_C(0^-) = 35 \text{ X } \frac{200}{200 + 500} = 10 \text{ V}$$

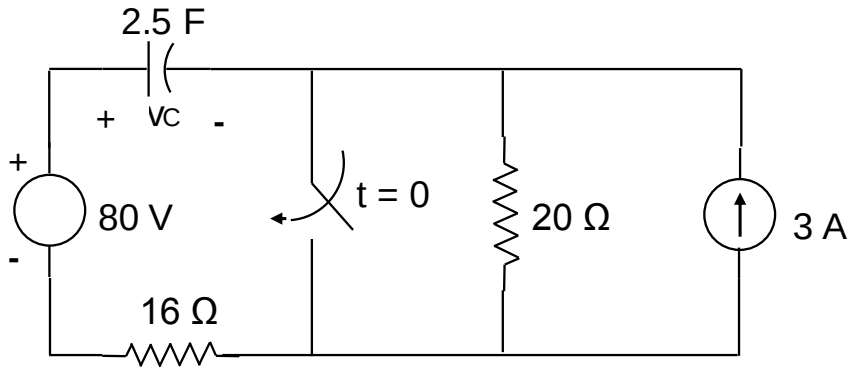
For time t > 0, capacitor voltage of 10 V is discharged through a resistor of 250 Ω.

Time constant RC = 250 X 2 X 10^{-3} = 0.5 s; $v_C(t) = 10\, e^{-2\,t}$ V

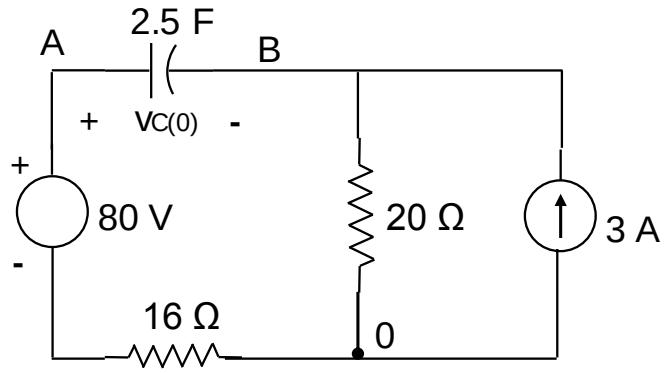
$$i_C(t) = C \frac{dv_C}{dt} = 2 \text{ X } 10^{-3} \text{ X } (-20)\, e^{-2\,t} \text{ A} = -40 \text{ X } 10^{-3}\, e^{-2\,t} \text{ A} = -0.04\, e^{-2\,t} \text{ A}$$

Thus $i(t) = -\, i_C(t) = 0.04\, e^{-2\,t}$ A

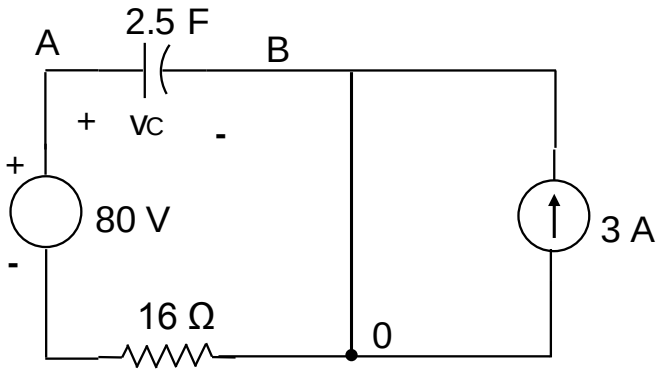
Example 7.10 Consider the circuit shown. The switch was in open position for a long time. It is operated as shown. Compute and plot the capacitor voltage for t > 0. Also find the time at which the capacitor voltage is 50 V.



Solution Circuit at time $t = 0^-$ is shown in Fig. (a).



(a)



(b)

Capacitor acts as open circuit. $I_{16\,\Omega} = 0$. Voltage $V_A = 80$ V and voltage $V_B = 60$ V

Thus $v_C(0) = 20$ V

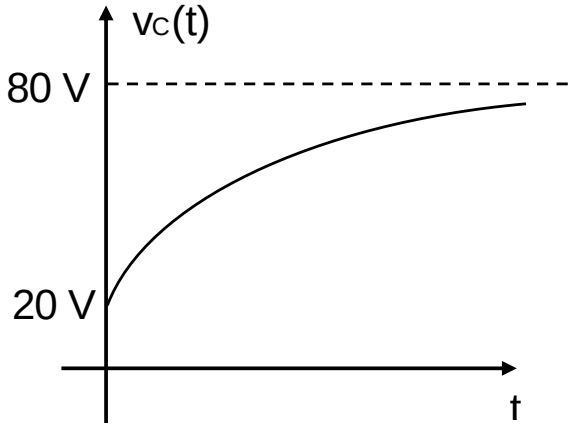
With the switch is in closed position, the circuit will be as shown in Fig. (b). With the steady state reached, Capacitor acts as open circuit. $I_{16\,\Omega} = 0$.

Voltage $V_A = 80$ V and voltage $V_B = 0$ V. Thus $v_C(\infty) = 80$ V

$RC = 16 \times 2.5 = 40$ s

Using $v_C(t) = v_C(\infty) + [v_C(0) - v_C(\infty)]\ e^{-\frac{1}{RC}t}$ we get

$v_C(t) = 80 + [20 - 80]\ e^{-0.025t} = 80 - 60\ e^{-0.025t}$ V



Plot of the capacitor voltage is shown.

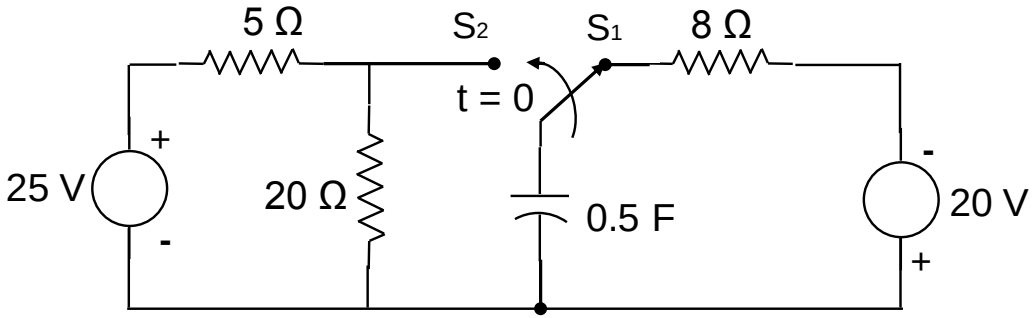
Let t_1 be the time at which the capacitor voltage = 50 V. Then

$80 - 60\ e^{-0.025t_1} = 50$ i.e. $60\ e^{-0.025t_1} = 30$ i.e. $e^{-0.025t_1} = 0.5$ i.e. $-0.025\ t_1 = -0.6932$

Thus $t_1 = 27.728$ s

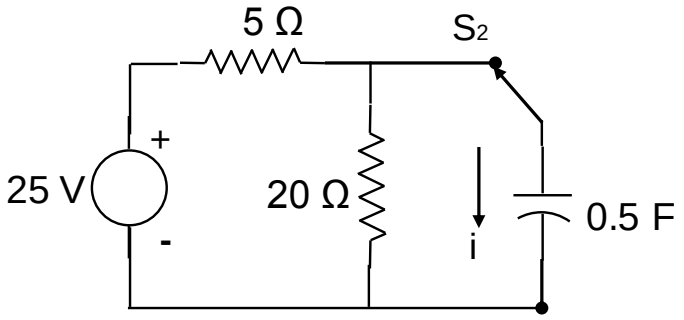
Capacitor voltage becomes 50 V at time $t_1 = 27.728$ s

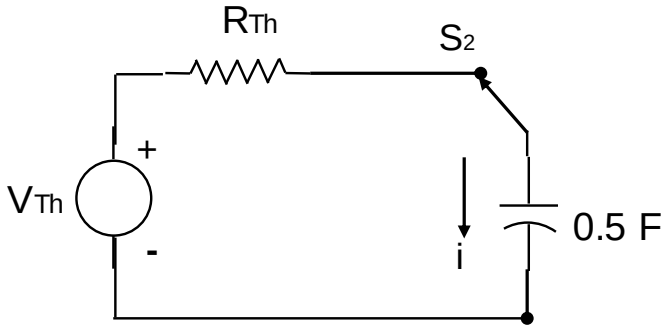
Example 7.11 Consider the circuit shown below. The switch was in position S_1 for a long time. It is operated as shown. Compute and plot the capacitor voltage for $t > 0$. Also find the time at which the capacitor voltage becomes zero.



Solution Voltage $v_C(0) = -20$ V

Circuit for time $t > 0$ and its Thevenin's equivalent are shown below.





$V_{Th} = \dfrac{20}{20+5} \times 25 = 20$ V $\qquad R_{Th} = 5 \,||\, 20 = 4\ \Omega$; Thus $RC = 4 \times 0.5 = 2$ s

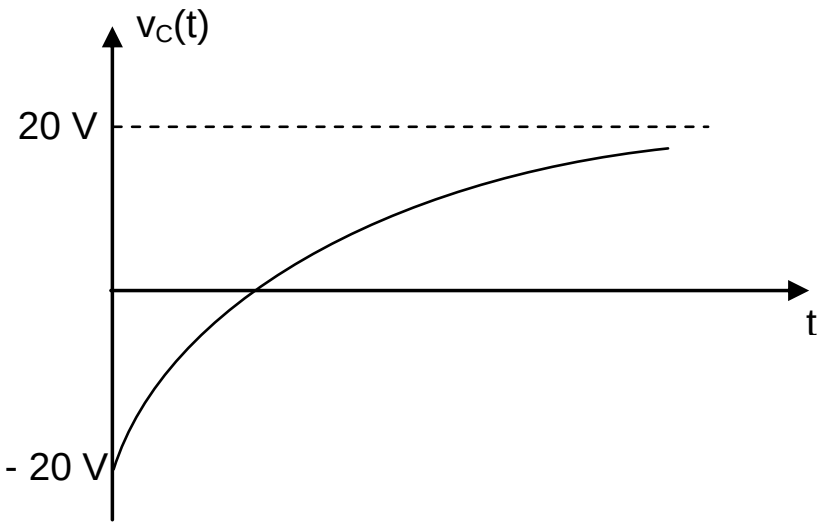
Using $v_C(t) = v_C(\infty) + [v_C(0) - v_C(\infty)]\ e^{-\frac{1}{RC}t}$ we get

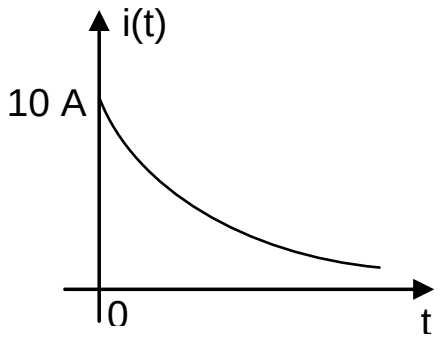
$v_C(t) = 20 + [-20 - 20]\ e^{-0.5t} = 20 - 40\ e^{-0.5t}$ V

$v_C(t) = 20 - 40\ e^{-0.5\,t}$ V

$i_C(t) = C\ \frac{dv_C}{dt} = 0.5 \text{ X } 20\ e^{-0.5\,t}$ A $= 10\ e^{-0.5\,t}$ A

Wave shapes of $v_C(t)$ and $i_C(t)$ are shown below.





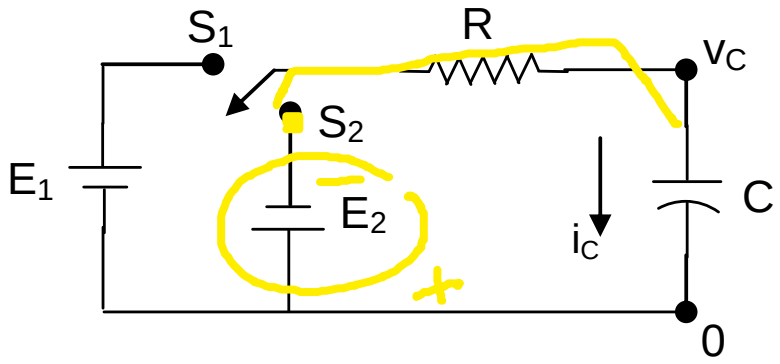
Let t_1 be the time at which the capacitor voltage reaches zero value. Then

$20 - 40\ e^{-0.5t_1} = 0$; i.e. $e^{-0.5t_1} = 0.5$; i.e. $-0.5\ t_1 = -0.6931$; Thus $t_1 = 1.3863$ s

Capacitor voltage reaches zero value at time $t_1 = 1.3863$ s

So far we have done transient analysis for one switching period. Now we shall illustrate how to carry out transient analysis for two switching period through an example.

Example 7.12 In the initially relaxed RC circuit shown the switch is closed on to position S_1 at time t = 0. After one time constant, the switch is moved on to position S_2. Find the complete capacitor voltage and current transients and show their wave forms.



E_1 = 20 V; E_2 = 40 V

R = 500 Ω

C = 0.5 μF

Solution RC = 500 X 0.5 X 10^{-6} s = 0.25 X 10^{-3} s = 0.25 ms 1/RC = 4000 s^{-1}

During the first switching period, capacitor gets charged from zero volt. Its voltage exponentially increases towards 20 V. Thus

vc= (E1-E1 e^-t\rc)

$$v_C(t) = 20\,(1 - e^{-4000\,t})\ V$$

At t = 1 time constant, $v_C = 20\,(1 - e^{-1}) = 12.64$ V

For the second switching operation, there is initial capacitor voltage of 12.64 V.

Let the second switching occurs at time $t' = 0$. Time $t' = 0$ implies time $t = 0.25 \times 10^{-3}$ s i.e. $t' = t - 0.25 \times 10^{-3}$. For $t' > 0$, capacitor voltage changes from its initial value, $v_C(0)$, of 12.64 V to final value, $v_C(\infty)$, of - 40 V. Knowing that

$$v_C(t) = v_C(\infty) + [v_C(0) - v_C(\infty)]\ e^{-\frac{1}{RC}t}$$ we get

$$v_C(t') = -40 + [12.64 + 40]\ e^{-4000\,t'} = 52.64\ e^{-4000\,t'} - 40 \text{ V}$$

Therefore, capacitor voltages for the two switching periods are

$$v_C(t) = 20\ (1 - e^{-4000\,t}) \text{ V} \quad \text{for } t > 0 \text{ and } \leq 0.00025 \text{ s}$$

$$v_C(t) = 52.64\ e^{-4000\,(t - 0.00025)} - 40 \text{ V for } t \geq 0.00025 \text{ s}$$

with $v_C(0.00025^-) = v_C(0.00025^+) = 12.64$ V

(Note that the capacitor voltage shall maintain continuity)

Knowing that

$v_C(t) = 20\,(1 - e^{-4000\,t})$ V for $t > 0$ and ≤ 0.00025 s

$v_C(t) = 52.64\, e^{-4000\,(t - 0.00025)} - 40$ V for $t \geq 0.00025$ s

For the first switching period

Capacitor current $i_C(t) = C \dfrac{dv_C}{dt} = 0.5 \times 10^{-6} \times 20 \times 4000\, e^{-4000\,t} = 0.04\, e^{-4000\,t}$ A

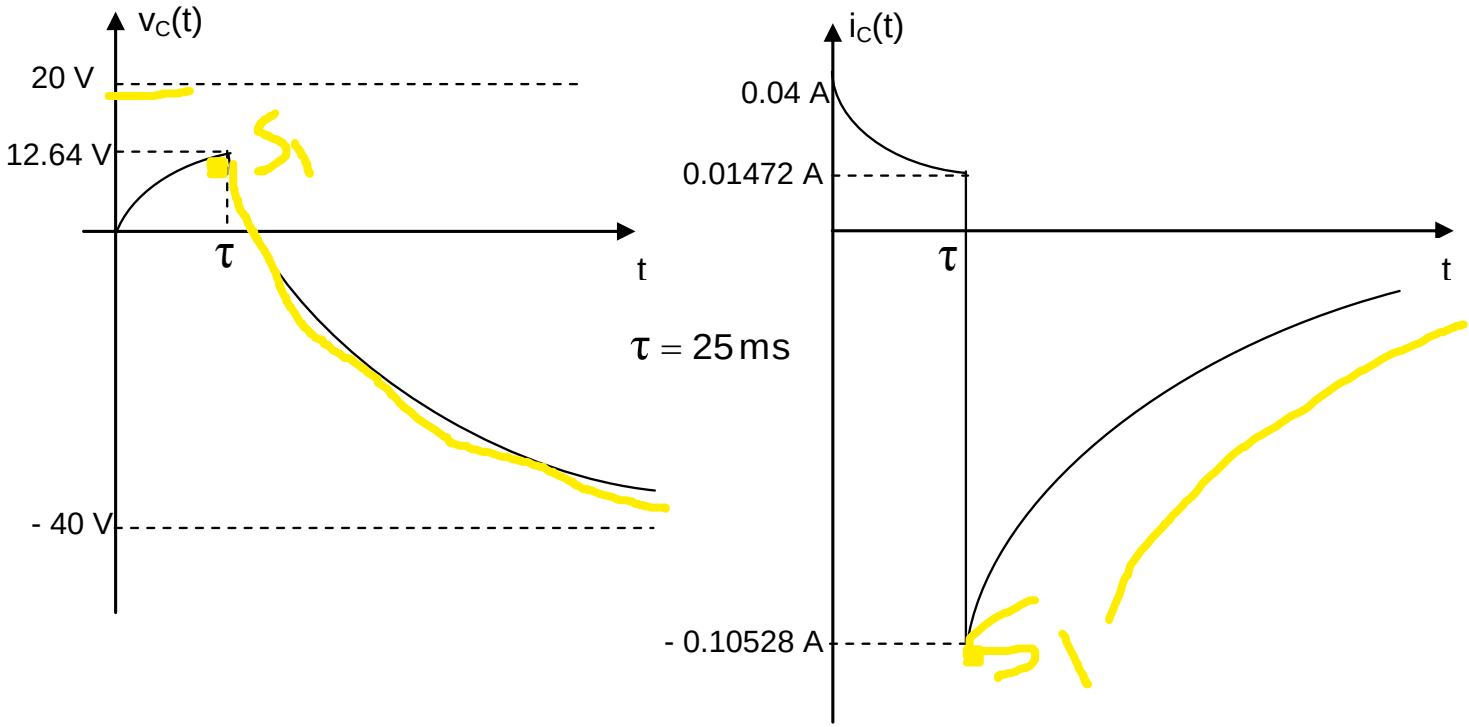
$i_C(0.00025^-) = 0.04\, e^{-1} = 0.01472$ A

For the second switching period,

$v_C(t') = 52.64\, e^{-4000\,t'} - 40$ V

$i_C(t') = 0.5 \times 10^{-6} \times (-52.64 \times 4000\, e^{-4000\,t'}) = -0.10528\, e^{-4000\,t'}$ A

i.e. $i_C(t - 0.00025) = -0.10528\, e^{-4000\,(t - 0.00025)}$ A $\qquad i_C(0.00025^+) = -0.10528$ A



Note: At the switching time, voltage across the capacitor does not have discontinuity i.e. $v_C(0.25 \text{ X } 10^{-3})^- = v_C(0.25 \text{ X } 10^{-3})^+$. On the other hand, the current through the capacitor has discontinuity at the instant of switching. The current just before switching and just after switching can be calculated by considering the circuit conditions at the respective time. At time $t = (0.25 \text{ X } 10^{-3})^-$, current $i = \dfrac{20 - 12.64}{500} = 0.01472$ A

At time $t = (0.25 \text{ X } 10^{-3})^+$, current $i = \dfrac{-40 - 12.64}{500} = -0.10528$ A

RC Circuit	RL Circuit
$\tau = RC \qquad \alpha = 1 / RC$	$\tau = L / R \qquad \alpha = R / L$
Switching at t = 0 $\quad v_C(0^+) = v_C(0^-)$	Switching at t = 0 $\quad i_L(0^+) = i_L(0^-)$
With DC, at SS capacitor acts as open circuit	With DC, at SS inductor acts as short circuit
$v_C(0) \neq 0;\ v_C(\infty) = 0;$ Then $v_C(t) = v_C(0)\ e^{-\frac{1}{RC}t}$	$i_L(0) \neq 0;\ i_L(\infty) = 0;$ Then $i_L(t) = i_L(0)\ e^{-\frac{R}{L}t}$
$v_C(0) = 0;\ v_C(\infty) \neq 0;$ Then $v_C(t) = v_C(\infty)\ (1 - e^{-\frac{1}{RC}t})$	$i_L(0) = 0;\ i_L(\infty) \neq 0;$ Then $i_L(t) = i_L(\infty)\ (1 - e^{-\frac{R}{L}t})$
$v_C(0) \neq 0;\ v_C(\infty) \neq 0;$ Then $v_C(t) = v_C(\infty) + [v_C(0) - v_C(\infty)]\ e^{-\frac{1}{RC}t}$	$i_L(0) \neq 0;\ i_L(\infty) \neq 0;$ Then $i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)]\ e^{-\frac{R}{L}t}$
$i_C(t) = C\frac{dv_C(t)}{dt}$	$v_L(t) = L\frac{di_L(t)}{dt}$